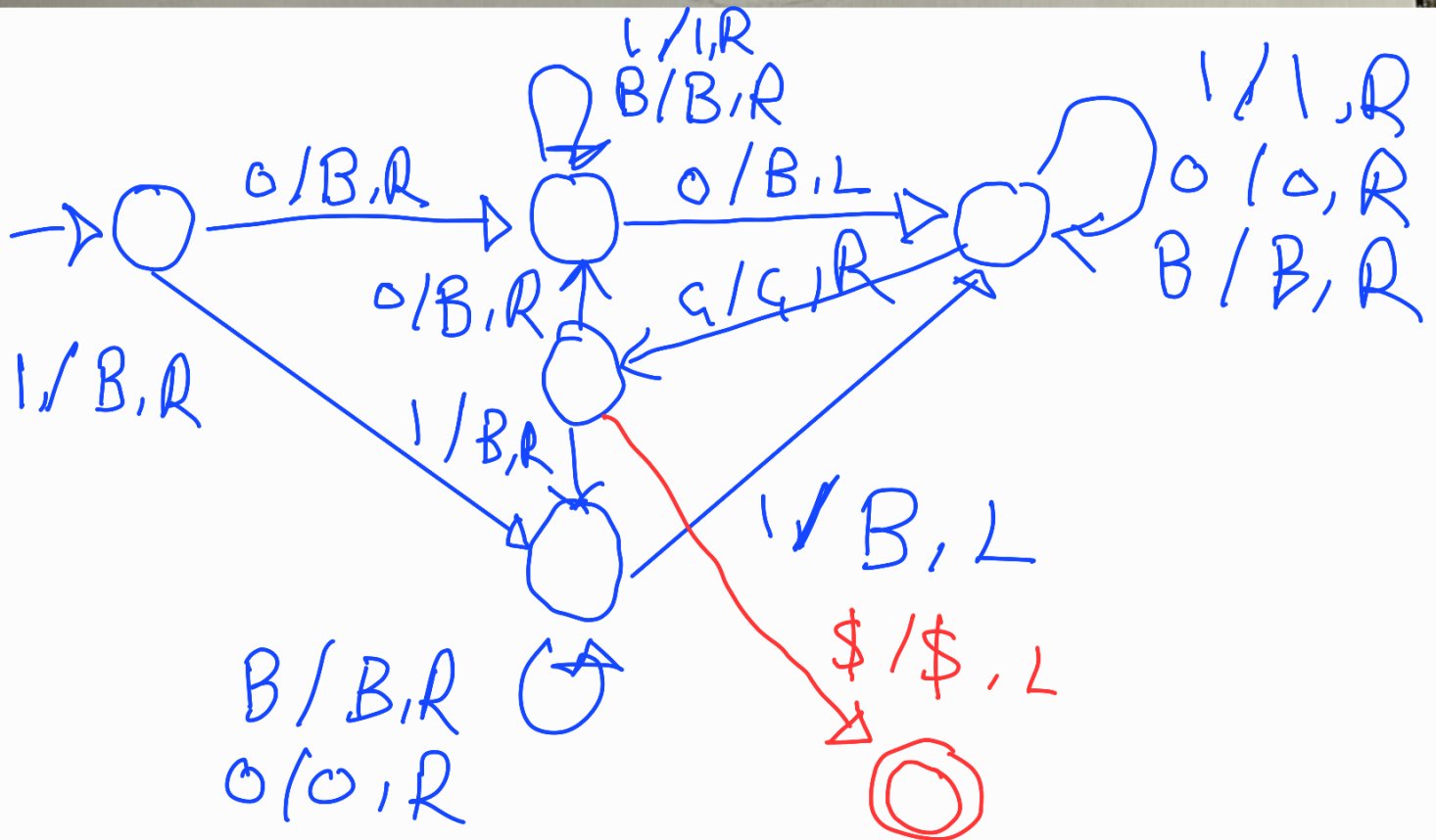
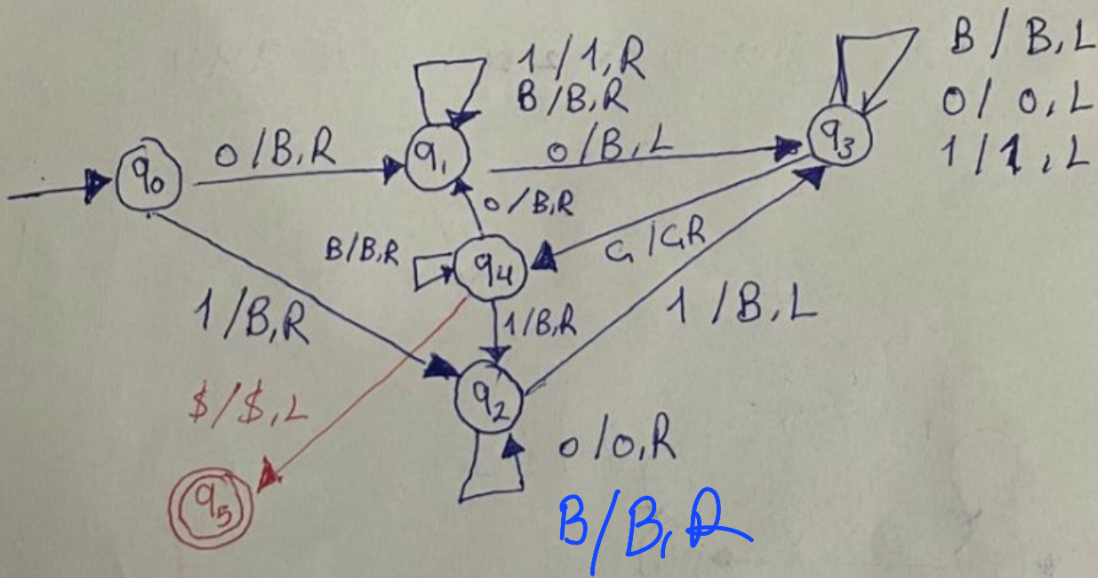


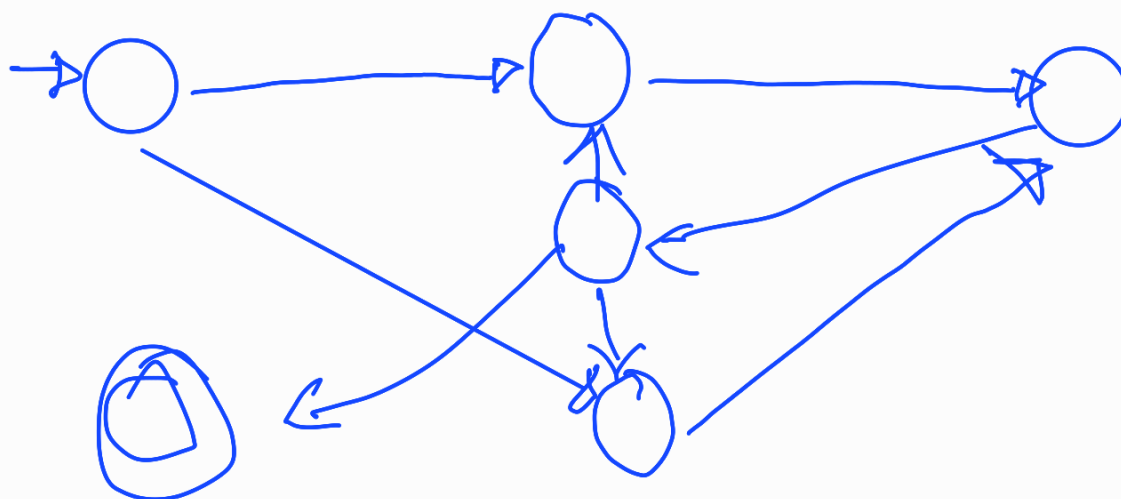
Örnek: (Turing Makinası)

[4]

$\Sigma = \{0, 1\}$ $|w| \geq 2$ çift sayıda "0" ve çift sayıda "1" içeren turing makinası tasarlayınız.

Başlangıç B B B B B B B B → Son
 B 1 1 0 1 0 0 1 0 \$ B





Örnek: $L = \{ (aa)^n (cc)^m b^k c b^k \mid n \geq 1, m \geq 0, k \geq 1 \}$

Bu dili tanıyan grameri tasarlayınız.

Bu grameri CNF'ye dönüştürünüz.

Çözüm: $V_N = \{S, D, E\}$, $V_T = \{a, b, c\}$

P: $S \Rightarrow aaS \mid aaD$

$D \Rightarrow ccD \mid bEb$

$E \Rightarrow bEb \mid c$

$T_a \Rightarrow a$

$T_b \Rightarrow b$

$T_c \Rightarrow c$

$S \Rightarrow aaS$ yerine

$S \Rightarrow T_a T_a S$

$S \Rightarrow T_a X$

$X \Rightarrow T_a S$

$S \Rightarrow aaD$ yerine

$S \Rightarrow T_a T_a D$

$S \Rightarrow T_a Y$

$Y \Rightarrow T_a D$

$D \Rightarrow ccD$ yerine

$D \Rightarrow T_c T_c D$

$D \Rightarrow T_c Z$

$Z \Rightarrow T_c D$

$D \Rightarrow bEb$ yerine

$D \Rightarrow T_b E T_b$

$D \Rightarrow T_b K$

$K \Rightarrow E T_b$

$E \Rightarrow bEb$ yerine

$E \Rightarrow T_b E T_b$

$E \Rightarrow T_b L$

$L \Rightarrow E T_b$

$S \Rightarrow T_a X \mid T_a Y$

$X \Rightarrow T_a S$

$Y \Rightarrow T_a D$

$D \Rightarrow T_c Z$

$Z \Rightarrow T_c D$

$D \Rightarrow T_b K$

$K \Rightarrow E T_b$

$E \Rightarrow c$

$E \Rightarrow T_b L$

$L \Rightarrow E T_b$

$T_a \Rightarrow a$

$T_b \Rightarrow b$

$T_c \Rightarrow c$

